

CMPE 491

Senior Design Project I

Deep Learning-Based Automated Visual Dataset Platform

Project Proposal

Team Members:

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Name of the project: Deep Learning-Based Automated Visual Dataset Platform

The URL of the project web page: https://visual-dataset-platform.github.io/deep-learning-based-automated-visual-dataset-platform/

Names of the supervisor: Mert Burkay Çöteli

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Project Description:

Preparing high-quality visual datasets for artificial intelligence and deep learning applications is often a time-consuming process that requires manual image collection, filtering, classification, and labeling. This project aims to develop an AI-powered web platform that automates visual dataset creation and reduces the manual effort required for data preparation. Users will be able to specify their desired image categories, dataset size, and filtering preferences through a user-friendly interface. The system will automatically collect relevant images from online sources, evaluate their quality and semantic relevance, and organize them into structured datasets suitable for machine learning applications.

Additionally, by combining automatic image collection and dataset management in a single platform, the project aims to make dataset creation easier and faster. The platform will provide a simple and user-friendly solution for researchers, students, and developers working on computer vision applications.